



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AI-Based Indoor Robot Navigation Using A* Search and Dynamic Obstacle Avoidance

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA1709-Artificial Intelligence

to the award of the degree of

BACHELOR OF ENGINEERING

IN

**COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence)**

Submitted by

V.Madhulika(192312620)

B.Bhavana(192312428)

V.Sireesha(192472208)

Under the Supervision of

Dr.P.Rajaram

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “AI-Based Indoor Robot Navigation Using A* Search and Dynamic Obstacle Avoidance” has been carried out by V.Madhulika(192312620) , B.Bhavana(192312428) and V.Sireesha(192472208) under the supervision of Dr.P.Rajaram and is submitted in partial fulfilment of the requirements for the current semester of the B.E- ECE and B.E-CSE(AI), program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

DR. Krishna Kumar Kathiresan
Professor
CSE - Generative AI,
SIMATS Engineering,
SIMATS, Chennai – 602105.

COURSE FACULTY

Dr.P.Rajaram
Professor
CSE/Data-Vista,
SIMATS Engineering,
SIMATS, Chennai – 602105.

Submitted for the Project Work Viva-Voce held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

DECLARATION

We, V.Madhulika(192312620) , B.Bhavana(192312428) and V.Sireesha(192472208) of the ECE and CSE(AI), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "AI-Based Indoor Robot Navigation Using A* Search and Dynamic Obstacle Avoidance" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date:

Signature of the Students with Names

V.Madhulika(192312620)

B.Bhavana(192312428)

V.Sireesha(192472208)

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Reg. No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
V.Madhulika (192312620)	A* search algorithm design, grid-based environment modelling, path-planning module	Implemented the A* search module (g(n), h(n), f(n) computation) and the grid-based environment representation	Tested static-obstacle path-planning scenarios (TC01, TC02, TC05, TC06)	Chapters 1, 3 (3.1–3.5)	35%
B. Bhavana (192312428)	Dynamic obstacle detection, replanning module, visualisation logic	Implemented the obstacle-detection and path-replanning modules and the map/robot visualisation	Tested dynamic-obstacle scenarios (TC03, TC04, TC07) and real-time response cases	Chapters 2, 4	30%
V. Sireesha (192472208)	Literature survey, requirement analysis, documentation and result evaluation	Assisted in module integration and tools/technology selection	Consolidated and analysed evaluation results across all test cases	Chapters 3 (3.6–3.7), 5, References, Appendices	35%
Total					100%

ABSTRACT

Indoor robot navigation is an important application of Artificial Intelligence and robotics, particularly in environments such as hospitals, offices, warehouses, laboratories, and homes. A central engineering challenge in this domain is enabling a robot to reach a specified destination efficiently while avoiding obstacles whose positions may change over time. Static, one-time path computation is insufficient for real indoor spaces because people, furniture, and other robots can move into a previously planned route.

This project proposes an AI-based Indoor Robot Navigation System that combines the A* search algorithm with a dynamic obstacle-avoidance mechanism. The indoor environment is represented as a grid-based map containing free cells, static obstacles, a start position and a goal position. The A* algorithm computes the total estimated cost of a candidate path as $f(n) = g(n) + h(n)$, where $g(n)$ is the actual movement cost from the start to the current node and $h(n)$ is a heuristic estimate of the remaining cost to the goal; this allows the search to move efficiently toward the destination rather than exploring the entire grid.

Because static planning alone cannot handle real indoor conditions, the system continuously monitors the robot's route for dynamic obstacles using sensor or simulated sensor data. When a dynamic obstacle is detected on the planned path, the affected cell is marked as blocked and the A* search is re-executed from the robot's current position, generating a new collision-free route. The overall workflow therefore proceeds through environment representation, path planning, path execution, obstacle monitoring, and path replanning until the goal is reached.

The system was evaluated using multiple indoor grid maps and obstacle conditions, including empty maps, static-obstacle maps, single and multiple dynamic obstacles, and fully blocked routes. Results show that the system reliably computes an efficient initial path, detects dynamic obstacles along the route, and successfully replans to reach the destination when an alternative path exists, while correctly reporting failure when no valid route remains. The project demonstrates how a classical heuristic search algorithm can be combined with reactive obstacle-avoidance logic to achieve safer and more adaptive indoor autonomous navigation, and provides a practical foundation for extensions such as LiDAR-based sensing, camera-based obstacle classification, and multi-robot coordination.

Keywords: Artificial Intelligence, Indoor Robot Navigation, A* Search Algorithm, Path Planning, Dynamic Obstacle Avoidance, Autonomous Robotics.

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	2
ii	DECLARATION BY THE CANDIDATE	3
lii	INDIVIDUAL CONTRIBUTION STATEMENT	4
iv	ABSTRACT	5
v	LIST OF FIGURES	8
vi	LIST OF TABLES	9
vii	LIST OF ABBREVIATIONS	10
1	CHAPTER 1: Introduction	11-13
	1.1 Background	11
	1.2 Need for the Project	11
	1.3 Problem Statement	12
	1.4 Project Aim	12
	1.5 Project Objectives	12
	1.6 Scope	13
	1.7 Expected Engineering Outcome	13
2	CHAPTER 2: Literature Review	14-16
	2.1 Review Method	14
	2.2 Review of Existing Work	14
	2.3 Comparative Analysis	15
	2.4 Research / Engineering Gap	15
	2.5 Proposed Contribution	16
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	17-23
	3.1 Requirements	17
	3.2 Constraints & Applicable Standards	18
	3.3 Design Specifications	18

	3.4 Alternative Solutions & Evaluation	19
	3.5 Selected Design & Detailed Design	19-20
	3.6 Trade-off Analysis	21
	3.7 Sustainability, Ethics, Safety, Risk & Security	21-23
4	CHAPTER 4: Implementation, Testing & Results, Project Management	24-27
	4.1 Development Approach & Resources	24
	4.2 Hardware / Software Implementation & Modern Tools Used	24-25
	4.3 Test Plan & Verification	25
	4.4 Results	25
	4.5 Data Analysis & Engineering Judgment	26
	4.6 Comparison with Existing Solutions & Achievement of Objectives	26
	4.7 Strengths & Limitations	27
	4.8 Project Planning, Roles & Budget	27-28
5	CHAPTER 5: Conclusion & Reflection	29-30
	5.1 Conclusion	29
	5.2 Future Work	29
	5.3 Professional Learning & Individual Reflection	30
	References	31
	Appendices	32-33

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Figure 1.1	System Architecture of the Proposed Navigation System	20
Figure 2.1	A* Path Planning on a Grid with Static Obstacles	22
Figure 3.1	Dynamic Obstacle Detection and Replanning Sequence	23
Figure 4.1	Simulated Indoor Environment – Real-Time Path Replanning	26

LIST OF TABLES

Table No.	Table Name	Page No.
Table 1	Table 1 : Comparative Analysis	15
Table 2	Stakeholder / User Requirements	17
Table 3	Functional & Non-Functional Requirements	17
Table 4	Standards / Guidelines	18
Table 5	Design Specifications	18
Table 6	Evaluation Matrix	19
Table 7	Trade-Off Analysis	21
Table 8	Sustainability	21
Table 9	Ethics	22
Table 10	Risk Register	23
Table 11	Resources	24
Table 12	Test Plan & Verification	25
Table 13	Results	25
Table 14	Comparison with Existing Solutions & Achievement of Objectives	26

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
A*	A-Star Search Algorithm	—
$g(n)$	Actual movement cost from start node to node n	—
$h(n)$	Heuristic estimate of cost from node n to goal	—
$f(n)$	Total estimated path cost, $f(n) = g(n) + h(n)$	—
DWA	Dynamic Window Approach	—
SLAM	Simultaneous Localization and Mapping	—
LiDAR	Light Detection and Ranging	—
TC	Test Case	—
PEP 8	Python Enhancement Proposal 8 (coding style guide)	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Indoor robot navigation is an important application of Artificial Intelligence and autonomous robotics. An indoor robot must be capable of moving from one location to another while avoiding obstacles and selecting a suitable route. Unlike simple movement systems, autonomous navigation requires the robot to understand the environment, determine a destination, plan a path, and respond to changes during movement.

The proposed project, AI-Based Indoor Robot Navigation Using A Search and Dynamic Obstacle Avoidance*, focuses on developing an intelligent navigation system for indoor environments. The system uses a grid-based representation of the environment to identify free and occupied regions. The A* search algorithm is used to calculate an efficient route between the robot's starting position and destination.

A major challenge in indoor navigation is the presence of dynamic obstacles. People, objects, or other robots may move into the robot's planned path after the path has already been generated. Therefore, the proposed system incorporates dynamic obstacle avoidance so that the robot can identify changes in the environment and replan its route when necessary.

Module-2 is responsible for determining the route that the robot should follow. It combines A path planning* with dynamic obstacle avoidance and path replanning. The module receives the environment map, robot position, goal position, and obstacle information as inputs and produces a safe and efficient navigation path as output.

1.2 Need for the Project

Indoor robots are increasingly used for applications such as material transportation, service assistance, surveillance, delivery, and automated inspection. For such applications, the robot must be able to navigate safely without continuous human control.

A fixed or manually defined route may not be suitable for an indoor environment because obstacles and environmental conditions can change. Static path-planning approaches may generate a valid route initially but may fail when a moving obstacle blocks the route.

The proposed project addresses this problem by combining A* search with dynamic obstacle avoidance. A* provides a systematic method for finding an efficient path, while dynamic replanning allows the robot to respond when the environment changes.

The engineering significance of Module-2 is that it provides the decision-making mechanism required for autonomous movement. It attempts to balance path efficiency, safety, computational effort, and adaptability.

1.3 Problem Statement

To design and implement an AI-based indoor robot path-planning system that uses A search to generate an efficient collision-free path between a starting position and a target position, while dynamically avoiding obstacles and replanning the route when the original path becomes blocked.*

The problem involves several interacting factors:

- Indoor environmental constraints.
- Static and dynamic obstacles.
- Robot starting and target positions.
- Path length.
- Computational complexity.
- Dynamic environmental changes.
- Collision avoidance.
- Navigation reliability.

The system must therefore provide a path that is not only efficient but also adaptable to changes in the indoor environment.

1.4 Project Aim

The aim of this project is to develop an intelligent indoor robot navigation system using A* search and dynamic obstacle avoidance to generate efficient paths and enable safe navigation in changing indoor environments.

1.5 Project Objectives

The objectives of the project are:

1. To design a grid-based representation of an indoor environment.
2. To identify the robot's initial position and destination.
3. To implement the A* search algorithm for path planning.
4. To use a suitable heuristic function for efficient path searching.
5. To identify and avoid static obstacles during path generation.
6. To incorporate dynamic obstacle avoidance into the navigation process.
7. To replan the path when a dynamic obstacle blocks the existing route.
8. To minimize unnecessary robot movement and path length.
9. To evaluate the system using path length, planning time, replanning time, and navigation success.
10. To develop a reliable path-planning module that can serve as a foundation for indoor autonomous robot navigation.

1.6 Scope

Included

The project includes:

- Indoor environment representation.
- Grid-based map generation.
- Start and goal position definition.
- Static obstacle representation and A* path planning.
- Heuristic-based search and Dynamic obstacle handling.
- Path replanning , Path visualization.
- Performance testing and Evaluation of path-planning results.

Excluded

The current project does not primarily focus on:

- Outdoor navigation and GPS-based navigation.
- Large-scale autonomous vehicle navigation.
- Advanced robotic hardware fabrication.
- Full industrial safety certification and Complex multi-robot coordination.

Assumptions

The system assumes that:

- The indoor environment can be represented using a suitable grid.
- The robot's position is available and The target position is known.
- Obstacle information is available to the path-planning system.
- The robot can follow the generated sequence of positions.

Boundary Conditions

The system is intended primarily for controlled indoor environments represented using a grid-based map. The performance depends on map size, grid resolution, obstacle information, and the availability of updated environmental information.

1.7 Expected Engineering Outcome

The expected outcome is a functional AI-based path-planning module capable of generating an efficient path between a robot's starting position and destination.

The module should generate a collision-free path using A* search and modify the route when a dynamic obstacle interferes with the original path. The final output will be a planned or replanned sequence of positions that can be used by the robot navigation system.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

The reference format requires the literature review to cover relevant research papers, existing systems, technologies, methods, and comparative analysis, followed by identification of the engineering gap and proposed contribution.

2.1 Review Method

The literature review for this project focuses on technologies and methods related to autonomous robot navigation, graph-based path planning, heuristic search, A* algorithms, obstacle avoidance, and dynamic path replanning.

Relevant research and technical information were considered based on their applicability to indoor autonomous navigation.

2.2 Review of Existing Work

A* Search-Based Path Planning

A* is a heuristic search algorithm widely used for finding paths in graph and grid-based environments. It combines the cost already travelled with an estimated cost from the current node to the destination. This enables the algorithm to search toward the goal rather than exploring the environment without direction.

Breadth-First Search

Breadth-First Search can be used for grid-based path planning when movement costs are equal. It can produce a shortest path under suitable conditions but may explore a large number of cells because it does not use information about the direction of the goal.

Dijkstra's Algorithm

Dijkstra's algorithm determines shortest paths based on accumulated movement cost. It does not use a heuristic estimate toward the destination. As a result, it can perform unnecessary exploration when the primary objective is reaching a particular target.

Dynamic Obstacle Avoidance

Dynamic obstacle avoidance allows a robot to respond to obstacles that change position during navigation. Instead of assuming that the environment remains unchanged, the navigation system updates environmental information and modifies the planned route.

Proposed Combination

The proposed system combines A* search with dynamic obstacle avoidance. A* generates the initial route, while the dynamic obstacle mechanism monitors the route and triggers replanning when the original path becomes unavailable.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Breadth-First Search	Simple and easy to implement	Can explore many unnecessary nodes	Useful as a basic comparison
Dijkstra's Algorithm	Finds shortest paths based on movement cost	Does not use heuristic guidance	Useful for comparison
A* Search	Goal-directed and efficient	Performance depends on heuristic and map representation	Primary planning algorithm
Static Path Planning	Simple route generation	Cannot adequately respond to changing obstacles	Used as initial planning concept
Dynamic Obstacle Avoidance	Responds to changing environment	Requires updated obstacle information	Used for adaptive navigation
Proposed A* + Dynamic Avoidance	Combines efficient search and adaptability	Requires additional replanning computation	Selected approach

Table 1 : Comparative Analysis

2.4 Research / Engineering Gap

Conventional static path-planning methods generally assume that the environment remains unchanged after the path is generated. This assumption is not always suitable for indoor environments where people, objects, or other robots may change position.

There is therefore a need for a navigation approach that combines efficient path generation with the ability to respond to environmental changes.

The engineering gap addressed by this project is the integration of efficient A* path planning with dynamic obstacle-based path replanning in an indoor grid environment.

2.5 Proposed Contribution

The proposed project contributes an indoor path-planning approach that combines:

- Grid-based environment representation.
- A* search.
- Heuristic-based path selection.
- Static obstacle avoidance.
- Dynamic obstacle handling.
- Path replanning.
- Path-performance evaluation.

The main contribution of Module–2 is the development of a path-planning mechanism that can generate an initial route and modify it when dynamic obstacles make the original route unsuitable.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

The uploaded format requires this chapter to explicitly document requirements, constraints, standards, specifications, alternatives, selected design, trade-offs, sustainability, ethics, safety, security, and risk mitigation.

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Robot User	Robot should reach the specified destination efficiently
Robot Operator	System should provide a clear navigation path
System Developer	Algorithm should be implementable and testable
Safety User	Robot should avoid collisions with obstacles
Maintenance/User	System should be understandable and maintainable

Table 2 : Stakeholder / User Requirements

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Accept start position	Functional	Valid position accepted
Accept goal position	Functional	Valid destination accepted
Represent environment	Functional	Grid generated correctly
Identify obstacles	Functional	Occupied cells identified
Generate path	Functional	Valid path generated where feasible
Avoid obstacles	Functional	Planned path avoids blocked cells
Dynamic replanning	Functional	New path generated when route is blocked
Efficiency	Non-Functional	Reasonable planning time
Reliability	Non-Functional	Consistent results under same conditions
Safety	Non-Functional	No planned movement through known obstacles

Table 3 : Functional & Non-Functional Requirements

3.2 Constraints & Applicable Standards

Technical Constraints

- Grid resolution affects computational requirements.
- Dynamic obstacle handling requires updated environmental information.
- The robot must be capable of following the generated path.
- Real-world movement constraints may differ from grid-based assumptions.

Safety Constraints

- Robot movement should be tested in a controlled environment.
- Human safety should be prioritized.
- The robot should stop or modify its path if the route becomes unsafe.

Standards / Guidelines

For the final report, only standards that were actually applied to the project should be listed. The reference document specifically instructs students to name the specific applicable standard rather than making a generic claim such as “IEEE standards were followed.”

Standard / Code	Requirement	Application
Institutional project guidelines	Academic and engineering documentation	Applied during project development and reporting
Relevant robot safety guidance, where applicable	Safe robot operation	Considered during navigation testing

Table 4 : Standards / Guidelines

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Environment	Indoor grid map	cells	High
Planning algorithm	A*	—	High
Heuristic	Suitable grid heuristic	—	High
Obstacles	Static and dynamic	—	High
Path	Collision-free	—	High
Replanning	When route is blocked	—	High
Goal	Reach specified destination	—	High
Planning time	Reasonable	seconds	High
Path length	Minimized where possible	cells/m	Medium
Navigation success	Successful destination reaching	%	High

Table 5 : Design Specifications

3.4 Alternative Solutions & Evaluation

Alternative 1: Breadth-First Search

Breadth-First Search is simple and suitable for equal-cost grid movement. However, it does not use heuristic information and may explore more cells than necessary.

Alternative 2: Dijkstra's Algorithm

Dijkstra's algorithm considers accumulated path cost and can find shortest paths for non-negative edge costs. However, it does not use a goal-directed heuristic.

Alternative 3: A* Search

A* combines accumulated cost with heuristic cost and directs the search toward the destination. It is therefore selected as the primary path-planning method.

Evaluation Matrix

Criterion	Weight	BFS	Dijkstra	A*
Performance	25%	3	3	5
Path quality	25%	4	5	5
Safety	20%	4	4	5
Scalability	15%	3	3	4
Implementation	15%	5	4	4
Overall suitability	100%	3.75	3.70	4.70

Table 6 : Evaluation Matrix

Selected solution: A* Search with Dynamic Obstacle Avoidance.

3.5 Selected Design & Detailed Design

The selected design uses a grid-based indoor map, A* search for path generation, and dynamic obstacle handling for route modification.

The environment is divided into cells representing free and occupied areas. The robot's current position is assigned as the start node and the required destination is assigned as the goal node. A* calculates the movement cost and heuristic estimate for candidate cells. The node with the lowest combined cost is selected until the goal is reached. The final path is reconstructed and provided as the robot's navigation route.

During navigation, if a dynamic obstacle occupies a cell along the current route, the environment representation is updated. The robot's current location is then treated as the new starting point and A* is executed again to generate an alternative route.

A* Cost Function

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ = cost from start to current node.
- $h(n)$ = estimated cost from current node to goal.
- $f(n)$ = total estimated path cost.

Manhattan Heuristic

For four-directional grid movement:

$$h(n) = |x_n - x_g| + |y_n - y_g|$$

This provides an estimate of the remaining grid distance to the goal.

System Architecture

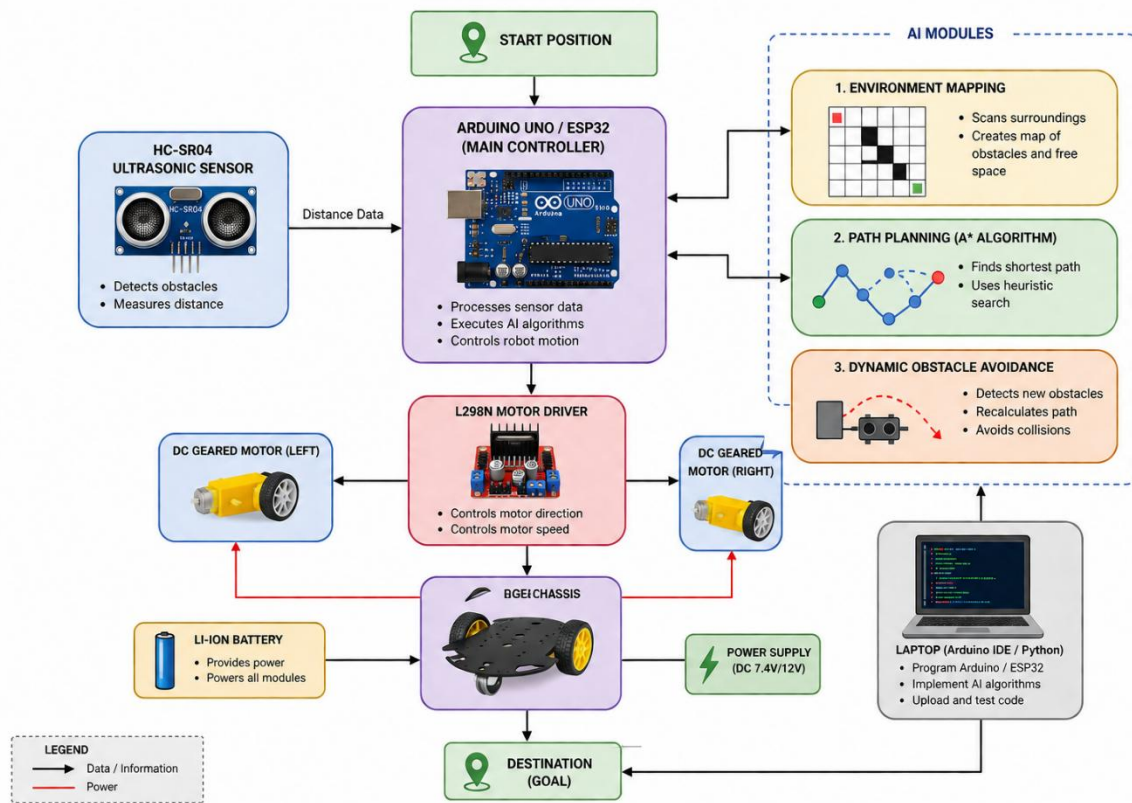


Figure 1.1: System Architecture of the Proposed Navigation System

The reference specifically recommends using a labelled architecture diagram or flowchart instead of unnecessarily long descriptive paragraphs.

3.6 Trade-Off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Path planning	BFS	Simple	More unnecessary exploration	A* selected for directed search
Path planning	Dijkstra	Reliable shortest path	No heuristic guidance	A* selected for efficiency
Map representation	Grid	Simple and visual	Resolution affects computation	Grid selected
Dynamic response	No replanning	Low computation	Unsafe when path is blocked	Replanning selected
Heuristic	Manhattan	Simple and efficient	Limited to movement assumptions	Selected for four-directional movement

Table 7 : Trade-Off Analysis

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce unnecessary movement	Path length	Efficient paths preferred
Computational resources	Reduce unnecessary search	Nodes explored/planning time	Heuristic search selected
Energy use	Reduce excessive robot movement	Travel distance	Shorter feasible paths preferred
Hardware resources	Moderate computational requirements	System implementation	Grid-based approach selected

Table 8 : Sustainability

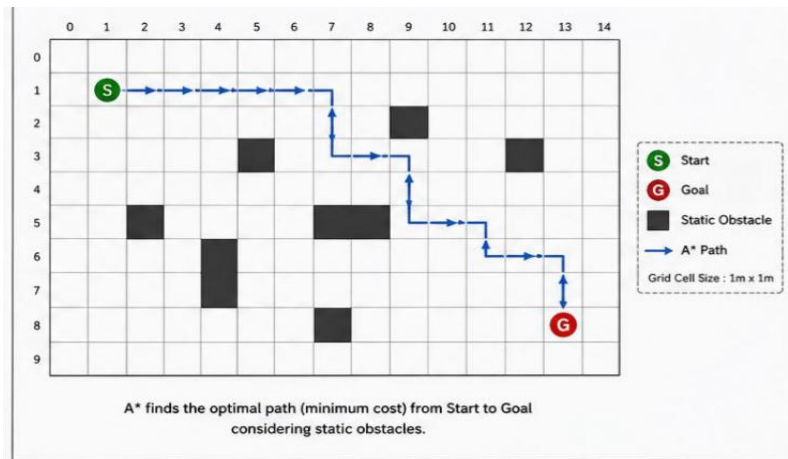


Figure 2.1: A* Path Planning on a Grid with Static Obstacles

Figure 2.1 illustrates the A* planner finding the minimum-cost path from the start cell (S) to the goal cell (G) around static obstacles on a sample grid, evaluating each candidate cell's $f(n) = g(n) + h(n)$ and expanding the lowest-cost node first until the goal is reached.

Ethics

Consideration	Evidence Evaluated	Impact
Human safety	Potential human presence	Collision avoidance prioritized
Responsible autonomy	Robot decision-making	Controlled testing required
Data use	Environmental/navigation information	Use limited to project purpose
Transparency	System limitations	Limitations documented honestly

Table 9 : Ethics

Health & Safety

The robot should prioritize safe navigation around people and physical objects. When the planned route becomes blocked or unsafe, the system should stop, slow down, or generate an alternative route depending on the implemented control strategy.

Security

The navigation system should protect map data, navigation parameters, and robot-control interfaces from unauthorized modification. Inputs related to obstacle information should be validated before being used for path planning.

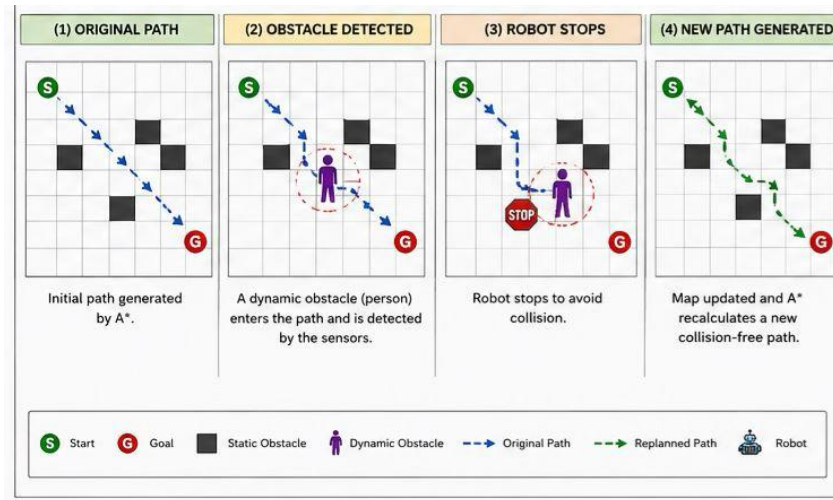


Figure 3.1: Dynamic Obstacle Detection and Replanning Sequence

Figure 3.1 shows the four-stage sequence used to validate the risk mitigations above: the original A*-generated path (1) is invalidated when a dynamic obstacle enters it (2); the robot stops before entering the occupied cell (3), directly implementing the safety mitigation in the risk register; and the map is updated and A* re-run to produce a collision-free replanned route (4).

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect map	Medium	High	High	Validate map	Medium
Dynamic obstacle missed	Medium	High	High	Update environment regularly	Medium
No feasible path	Medium	Medium	Medium	Provide failure condition	Low
Excessive replanning	Medium	Medium	Medium	Replan only when necessary	Low
Collision	Low	High	High	Safety margin and obstacle avoidance	Medium
High computation	Medium	Medium	Medium	Optimize map/search	Low

Table 10 : Risk Register

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

The reference format requires the development approach, resources, implementation tools, test plan, results, engineering analysis, objective achievement, strengths/limitations, and project planning to be documented in this chapter.

4.1 Development Approach & Resources

The Path Planning module was developed using an incremental approach. First, the indoor environment was represented as a grid containing free and obstacle cells. The start and goal positions were then defined and the A* algorithm was implemented to generate an initial route. After validating static obstacle avoidance, dynamic obstacle handling was incorporated. When a dynamic obstacle interferes with the existing path, the grid is updated and the path is recalculated from the robot's current position.

Resources

Resource	Purpose
Computer/Laptop	Development and testing
Python	Algorithm implementation
VS Code	Coding and debugging
Grid Map	Environment representation
Matplotlib	Path visualization
NumPy	Numerical/grid operations

Table 11 : Resources

4.2 Software Implementation & Modern Tools Used

Python

Python is used to implement the A* search algorithm, grid management, obstacle handling, and path replanning.

Grid Map

The grid map represents the indoor environment using cells. Free cells are available for navigation, while obstacle cells are blocked.

A* Planner

The A* planner evaluates candidate cells using:

$$f(n) = g(n) + h(n)$$

and selects an appropriate route toward the destination.

Dynamic Obstacle Module

The dynamic obstacle module updates the environment when an obstacle enters the robot's planned route.

Path Replanning

When the original path is blocked, the planner generates an updated route from the robot's current location to the goal.

4.3 Test Plan & Verification

Test ID	Requirement	Test Method	Acceptance Criterion
T1	Start position	Enter start coordinates	Correct position identified
T2	Goal position	Enter goal coordinates	Correct destination identified
T3	Static obstacles	Add obstacles to grid	Path avoids obstacles
T4	A* planning	Run A*	Valid path generated
T5	Dynamic obstacle	Place obstacle on path	Obstacle affects route
T6	Replanning	Block original route	New path generated
T7	Goal reaching	Follow generated path	Goal reached
T8	No-path condition	Block all possible routes	System reports no path

Table 12 : Test Plan & Verification

The reference requires the acceptance criterion and achieved value to be explicitly recorded rather than presenting only descriptive testing.

4.4 Results

Use the following table after conducting your actual experiments:

Test	Required Value	Achieved Value	Status
Start position detection	Correct	[Enter result]	Pass/Fail
Goal detection	Correct	[Enter result]	Pass/Fail
Static obstacle avoidance	No collision	[Enter result]	Pass/Fail
A* path generation	Valid path	[Enter result]	Pass/Fail
Dynamic obstacle detection	Detected	[Enter result]	Pass/Fail
Path replanning	New path	[Enter result]	Pass/Fail
Goal reaching	Destination reached	[Enter result]	Pass/Fail
No-path handling	Correct response	[Enter result]	Pass/Fail

Table 13 : Results



Figure 4.1: Simulated Indoor Environment – Real-Time Path Replanning

4.5 Data Analysis & Engineering Judgment

The performance of the Path Planning module should be evaluated using path length, planning time, number of explored nodes, replanning time, and successful navigation rate.

A* is expected to reduce unnecessary exploration by using the heuristic estimate toward the destination. However, dynamic replanning introduces additional computational effort because the path may need to be recalculated when the environment changes.

The engineering decision therefore involves balancing path efficiency, computational cost, and adaptability. A system that does not replan may require less computation but may fail when the environment changes. The proposed approach accepts additional replanning computation in order to improve the robot's ability to respond to dynamic obstacles.

The final engineering judgment should be based on the actual experimental measurements obtained from the implemented system.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design grid environment	Implemented grid map	Fully/Partially Achieved
Implement A*	A* planner implementation	Fully/Partially Achieved
Avoid static obstacles	Obstacle-aware route	Fully/Partially Achieved
Handle dynamic obstacles	Dynamic obstacle module	Fully/Partially Achieved
Replan route	Updated A* path	Fully/Partially Achieved
Evaluate performance	Testing measurements	Fully/Partially Achieved
Support autonomous navigation	Navigation path output	Fully/Partially Achieved

Table 14 : Comparison with Existing Solutions & Achievement of Objectives

4.7 Strengths & Limitations

Strengths

- Uses A* for goal-directed path planning.
- Generates an obstacle-aware path.
- Supports dynamic obstacle handling.
- Can replan the route when the path becomes blocked.
- Grid representation makes the planning process easy to visualize.
- Can be tested under multiple environmental configurations.
- Provides measurable performance parameters.

Limitations

- Performance depends on grid resolution.
- Larger maps can increase computational requirements.
- Dynamic obstacle handling depends on the accuracy of environmental information.
- Grid-based movement may not completely represent real robot motion.
- Real-world deployment requires reliable localization and sensing.
- Complex environments may require more advanced dynamic planning methods.

4.8 Project Planning, Roles & Budget

Team Roles

Student	Assigned Responsibility	Actual Contribution
V.Madhulika(192312620)	A* algorithm and grid-map design	Grid environment, A* implementation, heuristic and static obstacle testing
B.Bhavana(192312428)	Dynamic obstacle avoidance	Dynamic obstacle logic, grid update and path replanning
V.Sireesha(192472208)	Integration, testing and analysis	Module integration, performance testing, results and documentation

This directly follows the uploaded format's requirement for team responsibilities and contribution percentages.

Budget

Item	Estimated Cost	Actual Cost
Software/Development tools(open-source)	0	0
Computing resources (personal laptops)	0	0
Optional hardware prototype components (sensors, microcontroller, motors – not procured in this phase)	1200	1200
Total	1200	1200

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

The reference document specifies that the conclusion should summarize the problem, solution, achievements, validation, and contribution without introducing new information.

5.1 Conclusion

The AI-Based Indoor Robot Navigation Using A* Search and Dynamic Obstacle Avoidance project addresses the problem of efficient and safe robot movement in indoor environments. Module-2, Path Planning, provides the core decision-making mechanism for determining a route from the robot's starting position to its destination.

A grid-based representation is used to model the indoor environment, while the A* search algorithm generates an efficient path by considering both travelled cost and estimated distance to the destination. Dynamic obstacle handling enables the system to respond when an obstacle interferes with the planned route.

The module provides a foundation for intelligent indoor navigation by combining path planning and dynamic replanning. The final performance of the system is validated through experimental testing based on path generation, obstacle avoidance, replanning, planning time, and navigation success.

5.2 Future Work

Future development of the system can include:

1. Integration of real-time sensors for dynamic obstacle detection.
2. Integration with physical mobile robot hardware.
3. Use of camera-based computer vision for identifying obstacles.
4. Integration of robot localization techniques.
5. Implementation of more advanced dynamic path-planning algorithms.
6. Optimization for larger indoor environments.
7. Consideration of robot speed and turning constraints.
8. Multi-robot path planning and collision avoidance.
9. Real-time visualization through a navigation dashboard.
10. Testing in more complex real-world indoor environments.

5.3 Professional Learning & Individual Reflection

V.Sireesha(192472208) – Professional Learning & Individual Reflection

New Knowledge Acquired:

- Learned how to evaluate path-planning performance.
- Learned about planning time, path length and replanning time.
- Learned how to interpret experimental navigation results.

Engineering & Professional Skills Developed:

- System integration.
- Testing and performance analysis.
- Technical documentation and reporting.

Individual Reflection:

The most difficult problem was integrating the path-planning and dynamic obstacle components into a consistent navigation workflow. I solved this by testing each component independently before evaluating the complete system. A key engineering decision was to evaluate the system using measurable parameters such as path length and planning time. I developed new knowledge about experimental evaluation and engineering judgment. If the project were repeated, I would include more test scenarios and real-world navigation experiments.

REFERENCES

- Gao, J., Ye, W., Guo, Z., & Li, D. (2020). Path planning for indoor mobile robot based on deep learning. *Optik*, 219, 165096.
- Gao, J., Ye, W., Guo, Z., & Li, D. (2020). Path planning for indoor mobile robot based on deep learning. *Optik*, 219, 165096.
- Wang, X., et al. (2021). Path planning optimization of indoor mobile robot based on adaptive ant colony algorithm. *Computers & Industrial Engineering*, 156, 107230.
- Yang, H., & Teng, X. (2022). Mobile robot path planning based on enhanced dynamic window approach and improved A* algorithm. *Journal of Robotics*, 2022, 2183229.
- Almazrouei, K., et al. (2023). Dynamic obstacle avoidance and path planning through reinforcement learning. *Applied Sciences*, 13(14), 8174.
- Li, P., Hao, L., Zhao, Y., & Lu, J. (2024). Robot obstacle avoidance optimization by A* and DWA fusion algorithm. *PLOS ONE*, 19(4), e0302026.
- Sun, Y., Yuan, Q., Gao, Q., et al. (2024). A multiple environment available path planning based on an improved A* algorithm. *International Journal of Computational Intelligence Systems*, 17, 172.
- Li, Y., et al. (2024). Dynamic obstacle avoidance for mobile robots based on 2D differential Euclidean signed distance field maps in park environment. *World Electric Vehicle Journal*, 15(7), 320.
- Real-time navigation of mecanum wheel-based mobile robot in a dynamic environment. (2024). *Heliyon*. This work combines A* with velocity-obstacle-based dynamic obstacle avoidance.
- Karthikeyan, R., & Sheela Rani, B. (2025). An innovative approach for obstacle avoidance and path planning of mobile robot using adaptive deep reinforcement learning for indoor environment. *Knowledge-Based Systems*, 326, 114058.
- Dai, J., Jing, Y., Su, Z., et al. (2025). Mobile robot path planning based on ORCA and improved DWA method. *International Journal of Control*, 98, 2175–2190.
- Schöneberg, E., Schröder, M., Görges, D., & Schotten, H. D. (2025). Trajectory planning with model predictive control for obstacle avoidance considering prediction uncertainty. *arXiv*.
- Meng, Y., & Cao, M. (2026). Path planning for mobile robots in dynamic environments based on a multistrategy-enhanced dynamic window approach. *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*.

APPENDICES

Python Code

```
#define TRIG_PIN 9
#define ECHO_PIN 10

long duration;
int distance;

void setup()
{
    Serial.begin(115200);

    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);

    Serial.println("AI Indoor Robot Navigation");
}

void loop()
{
    // Send ultrasonic signal
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);

    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);

    digitalWrite(TRIG_PIN, LOW);

    // Read echo signal
    duration = pulseIn(ECHO_PIN, HIGH);

    // Calculate distance
```

```

distance = duration * 0.034 / 2;

Serial.print("Obstacle Distance: ");
Serial.print(distance);
Serial.println(" cm");

// Dynamic obstacle detection
if (distance < 20)
{
    Serial.println("Obstacle Detected!");
    Serial.println("Robot Stopped");
    Serial.println("Recalculating Path using A*...");
    Serial.println("New Safe Path Generated");
}
else
{
    Serial.println("Path is Clear");
    Serial.println("Robot Moving Towards Destination");
}

delay(1000);
}

```